

C2_03 Gate A Protected Review - Astra RC4

Retail Sales & Returns. Open only after a saved independent Gate A model attempt; close the review before changed repair proof.

1. Objective controls

Control	Expected	Why it matters
Fact rows	Sales 120 Returns 6	Separate fact grains; returns are not sales-line rows.
Product key	10 raw / 9 unique duplicate P04	One-side uniqueness fails even though the duplicate attributes happen to match.
Unknown product	P99 on 1 sales line gross 3,248.52	Inner-joining it away makes gross sales 297,902.61 instead of 301,151.13.
Sales control	60 orders / 120 lines gross 301,151.13	Proves order-line grain and independent monetary control.
Returns control	6 events 7,732.81	Keep return-event grain separate; subtraction needs an explicit business/date contract.
Date-role contrast	OrderDate rows Apr/May/Jun = 44/47/29; ShipDate = 39/47/34	Same sales facts move between months when the role changes.
Targets	12 Region-Month rows total 618,000	Higher-grain fact must filter through conformed dimensions, not repeat across sales lines.

2. Expected reasoning path

Begin with three grains: FactSales = one order line; FactReturns = one return event; FactMonthlyTargets = one Region-Month target. A direct fact-to-fact shortcut is not a star-schema substitute.

P04 is an exact duplicate in the supplied dirty product file. That makes a documented deterministic dedupe possible for this fixture, but the model must first prove why the duplicate is safe; M:M is not the repair.

P99 contributes 3,248.52 of gross sales. Dropping unmatched products can leave a professional-looking model while changing the controlled total. Preserve an unknown/exception path until ownership resolves the key.

OrderDate and ShipDate produce different monthly row/totals. The active/default date role must follow the report question; prove the alternate role with a tiny known filter case.

Region-Month targets are a higher-grain fact. Create/use a conformed Region and Date/Month path so a target is not multiplied by transaction lines. Reconcile target rows before any target-vs-actual measure.

3. Critical failures

- Uses M:M or Both to bypass duplicate P04.
- Drops P99 with an INNER JOIN or filter and still signs off gross sales.
- Joins Region-Month target rows to sales transaction rows and sums repeated target.
- Treats Returns as if it shared Sales line grain without an explicit key/date contract.
- Chooses OrderDate/ShipDate because a visual “looks right” without a boundary filter test.
- Releases without row/key/money reconciliation and exception evidence.

4. Competency floors

Competency	Required proof
Model contract / grain	Every fact and higher-grain table has an explicit row meaning; no mixed-grain shortcut.
Keys / star design	One-side uniqueness and orphan handling are evidenced before relationships; no arbitrary dedupe.
Relationships / date roles	Cardinality, active state and direction documented; tiny filter tests prove both required date roles.
Power Query / reconciliation	Traceable source-staging-model-ready path plus independent rows/keys/totals/exceptions.
Performance / handoff	Correctness first; runtime performance evidence only when actually observed; RELEASE/BLOCK and repair path are explicit.

Changed repair proof

Close this review. Repair only the named weak competency using a changed product/key/date/target case. Gate B is a different clinic model and opens only when explicitly assigned.